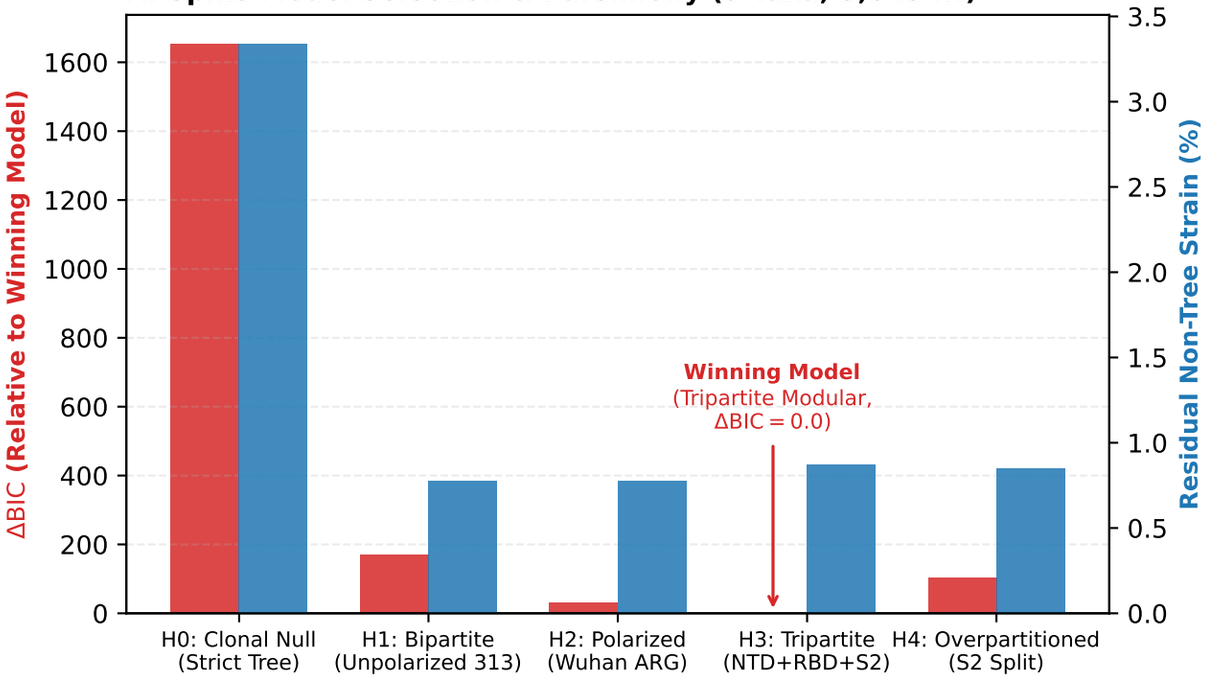
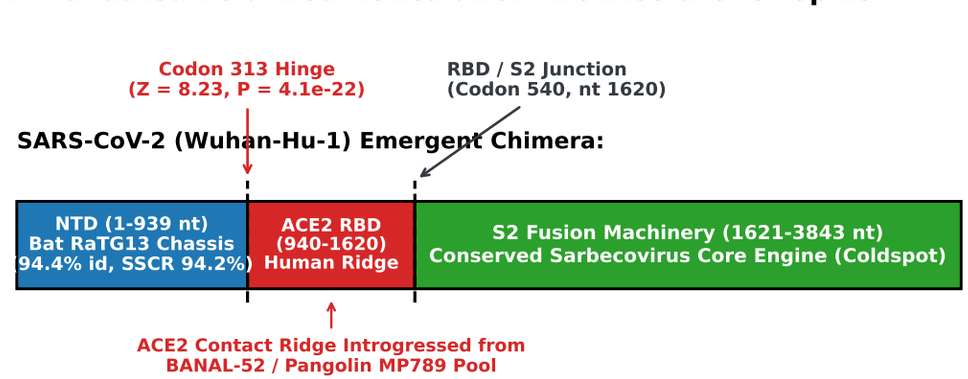
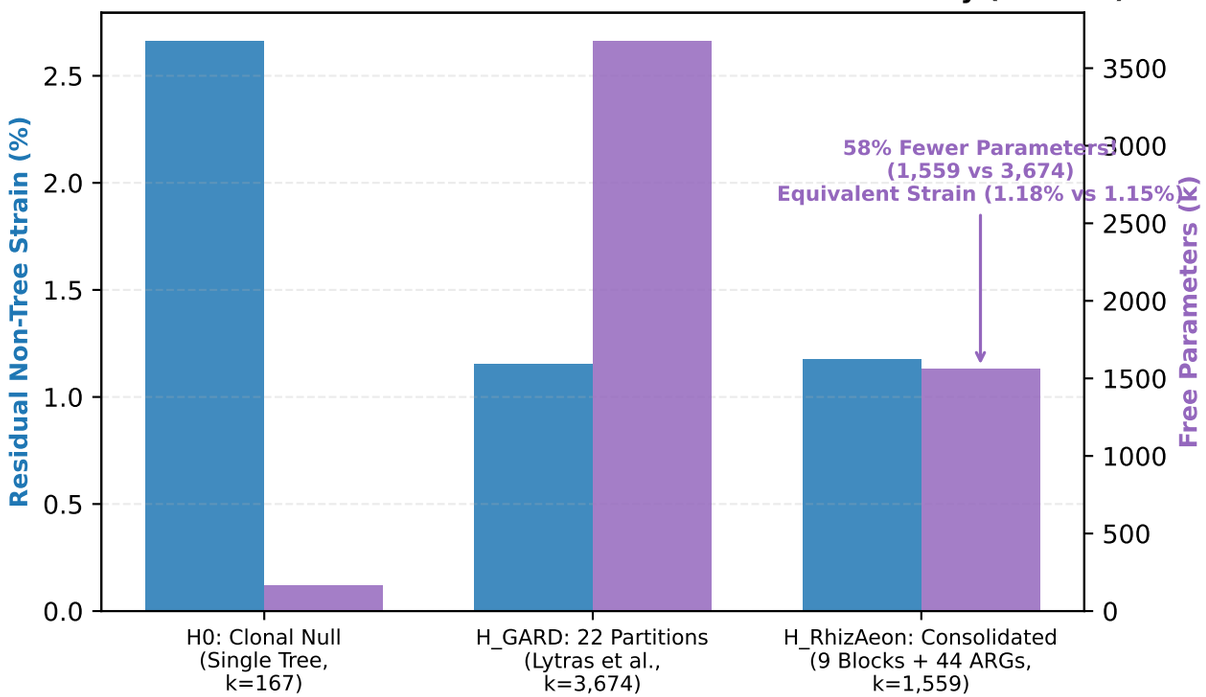
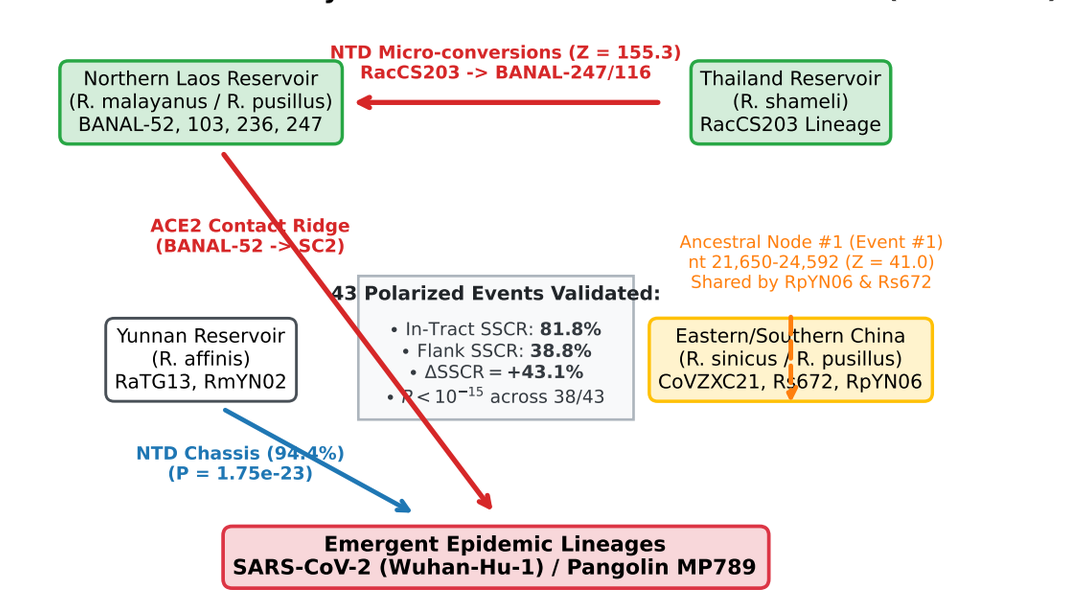

A. Spike Model Selection & Parsimony (9 Taxa, 3,843 nt)
ΔBIC (Relative to Winning Model)
Residual Non-Tree Strain (%)
Winning Model
(Tripartite Modular,
ΔBIC = 0.0)
H0: Clonal Null
(Strict Tree)
H1: Bipartite
(Unpolarized 313)
H2: Polarized
(Wuhan ARG)
H3: Tripartite
(NTD+RBD+S2)
H4: Overpartitioned
(S2 Split)


B. Validated Polarized Reticulation Architecture for Spike
Codon 313 Hinge
(Z = 8.23, P = 4.1e-22)
RBD / S2 Junction
(Codon 540, nt 1620)
SARS-CoV-2 (Wuhan-Hu-1) Emergent Chimera:
NTD (1-939 nt)
Bat RaTG13 Chassis
(94.4% id, SSCR 94.2%)
ACE2 RBD
(940-1620)
Human Ridge
S2 Fusion Machinery (1621-3843 nt)
Conserved Sarbecovirus Core Engine (Coldspot)
ACE2 Contact Ridge Introgressed from
BANAL-52 / Pangolin MP789 Pool
Statistical Validation of Polarized Transfer (History Engine Metrics):
• In-Tract Synapomorphic Concordance (SSCR): 94.2% match to Bat RaTG13
• Flank Synapomorphic Concordance: 62.1% (Polarization Contrast ΔSSCR = +32.1%)
• Polarity Significance (Fisher's Exact Test): P = 1.75 × 10−23 (Undeniable Reticulation Direction)
• Manifold Strain Collapse: Residual Strain drops from 3.34% → 0.87% (74.0% strain explained)


C. Whole-Genome Model Selection & Parameter Efficiency (83 Taxa)
Residual Non-Tree Strain (%)
Free Parameters (k)
58% Fewer Parameters
(1,559 vs 3,674)
Equivalent Strain (1.18% vs 1.15%)
H0: Clonal Null
(Single Tree,
k=167)
H_GARD: 22 Partitions
(Lytras et al.,
k=3,674)
H_RhizAeon: Consolidated
(9 Blocks + 44 ARGs,
k=1,559)


D. Macro-Evolutionary Sarbecovirus Reticulation Network (44 Events)
NTD Micro-conversions (Z = 155.3)
RacCS203 -> BANAL-247/116
Northern Laos Reservoir
(R. malayanus / R. pusillus)
BANAL-52, 103, 236, 247
Thailand Reservoir
(R. shameli)
RacCS203 Lineage
ACE2 Contact Ridge
(BANAL-52 -> SC2)
Ancestral Node #1 (Event #1)
nt 21,650-24,592 (Z = 41.0)
Shared by RpYN06 & Rs672
43 Polarized Events Validated:
• In-Tract SSCR: 81.8%
• Flank SSCR: 38.8%
• ΔSSCR = +43.1%
• P < 10−15 across 38/43
Yunnan Reservoir
(R. affinis)
RaTG13, RmYN02
Eastern/Southern China
(R. sinicus / R. pusillus)
CoVZXC21, Rs672, RpYN06
NTD Chassis (94.4%)
(P = 1.75e-23)
Emergent Epidemic Lineages
SARS-CoV-2 (Wuhan-Hu-1) / Pangolin MP789